

Lecture 10 Tutorial — Answers

Q1. AI System Pipeline

$$\boxed{E \rightarrow C \rightarrow A \rightarrow D \rightarrow B}$$

That is:

Formulate Problem $\rightarrow$ Collect Data $\rightarrow$ Feature Extraction $\rightarrow$ Machine Learning $\rightarrow$ System Deployed

Q2. Intelligence Category

1-A, 2-A, 3-C

1. Outlier detection $\rightarrow$ **A. Computation Intelligence**
2. Emotion detection $\rightarrow$ **A. Computation Intelligence**
3. Creative writing / story generation $\rightarrow$ **C. Creative Intelligence**

Q3. Computer Vision Tasks vs Applications

1-A, 2-B, 3-C, 4-D

1. Autonomous vehicles $\rightarrow$ **A. Object detection/recognition**
2. Medical imaging (tumour boundary delineation) $\rightarrow$ **B. Image segmentation**
3. Surveillance systems (tracking movement in a scene) $\rightarrow$ **C. Motion analysis**
4. Visual question answering over images $\rightarrow$ **D. Scene understanding (Q&A system)**

Q4. Traditional vs Deep Learning in Vision

1. **B.** Haar cascades
2. **A.** CNN
3. **B.** ViT

Q5. Tool Selection

1. **A.** OpenCV
2. **A.** MMDetection
3. **A.** Kaldi

Q6. Audio Tasks and Methods

1–A, 2–B, 3–C, 4–D, 5–E, 6–F

1. Speech feature extraction → **A. MFCCs**
2. Sequential audio modelling (classic approach) → **B. HMMs**
3. Source separation → **C. NMF**
4. Temporal dependencies for speech-to-text → **D. RNN/LSTM**
5. Multilingual ASR (modern approach) → **E. Transformer model (e.g., Whisper)**
6. High-fidelity music synthesis → **F. Diffusion models**

Q7. Audio Application Mapping

1. **A.** Speaker identification/verification
2. **A.** Environmental sound detection
3. **A.** Noise cancellation / sound enhancement

Q8. NLP Levels

1–Syntax-level, 2–Syntax-level, 3–Semantic-level, 4–Semantic-level, 5–Semantic-level

1. Tokenization / POS tagging → **Syntax-level**
2. Dependency parsing → **Syntax-level**
3. Named Entity Recognition (NER) → **Semantic-level**
4. Sentiment analysis → **Semantic-level**
5. Machine translation → **Semantic-level**

Q9. Traditional vs Deep Learning in NLP

1. Bag-of-Words (BoW) **A. Traditional**
2. TF-IDF weighting **A. Traditional**
3. Transformer models (BERT, GPT) **B. Deep Learning**
4. RNNs/LSTMs **B. Deep Learning**

Q10. Multi-modal Processing Pipeline

Multi-modal Input → Feature Extraction → (Text Features, Audio Features, Visual Features) → Fusion Layer → Decision

Q11. Multi-modal Core Challenges

The listed core challenges are:

- A. **A.** Cross-modal alignment (e.g., audio-visual synchronization)
- B. **B.** Feature fusion strategies
- C. **C.** Context-aware interpretation

The incorrect options are:

- D. **D.** Chi-square test for variance
- E. **E.** Laplacian correction

Q12. Application-to-Subfield-and-Task

1. Subfield D, Task Cross-modal alignment / feature fusion
2. Subfield B, Task Automatic Speech Recognition (ASR)
3. Subfield C, Task Text summarization
4. Subfield A, Task Object detection/recognition

More explicitly:

1. Automated lecture transcription with slide analysis
→ Subfield D (Multi-modal Media Processing), Task Cross-modal alignment / feature fusion
2. Real-time background noise suppression in video calls
→ Subfield B (Computer Auditory Systems), Task Automatic Speech Recognition (ASR)
3. A chatbot that answers questions and summarizes long documents
→ Subfield C (NLP), Task Text summarization
4. Traffic monitoring that detects vehicles in live video
→ Subfield A (Computer Vision), Task Object detection/recognition